

Answers:

1. True or False: Euler angles can suffer from gimbal lock.
 - a. True
2. True or False: Euler angles can be interpolated without issues.
 - a. False
3. True or False: It's usually preferable to use Euler angles over rotation matrices in user interfaces.
 - a. True
4. True or False: The order in which you apply rotations in Euler angles does not matter.
 - a. False
5. True or False: Euler angles are often easier for humans to understand and manipulate than rotation matrices.
 - a. True
6. True or False: All game environments apply Euler angle rotations in the same order.
 - a. False, you'll need to check what the default order is.
7. True or False: You can avoid gimbal lock by simply converting Euler angles to a rotation matrix afterwards.
 - a. False
8. True or False: Rotation matrices can be interpolated without issues.
 - a. False
9. True or False: Rotation matrices are more memory-efficient compared to Euler angles.
 - a. False, rotation matrices require 9 values (3x3 matrix), while euler angles only require 3 values.
10. True or False: In a rotation matrix, each row (or column) must be a unit vector.
 - a. True
11. What is gimbal lock? Provide a brief explanation.
 - a. Generally the idea is just to understand that you've essentially lost a degree of freedom in your rotations. In the example we showed in the video, the red and blue or x and z axis rotations were essentially the same rotation. So if you were to imagine it being able to look left and right, after it was gimbal locked, that wasn't possible anymore.
12. Describe a real-world scenario in a game where gimbal lock could cause issues.
 - a. An easy example would be a spaceship controller using Euler angles. If you were to dive forward by 90°, you could potentially gimbal lock, and that would completely restrict your ability to roll the ship.
13. Is gimbal lock an issue when using rotation matrices? Why or why not?
 - a. If you use rotation matrices directly, no it isn't an issue. But if you convert FROM Euler angles to a rotation matrix, say you're using Euler angles in code and then convert to a rotation matrix afterwards in the hopes of avoiding gimbal lock, that won't work. The problem is inherent to Euler angles, and once it's occurred, converting to a rotation matrix won't save you at that point.